

- 11-2 Find the driving point impedance seen by the voltage source in Figure P11-2 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

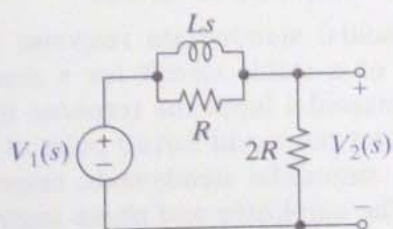


FIGURE P11-2

- 11-3 Find the driving point impedance seen by the voltage source in Figure P11-3 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

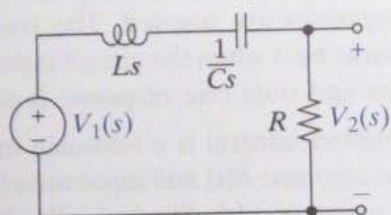


FIGURE P11-3

- 11-4 Find the driving point impedance seen by the voltage source in Figure P11-4 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

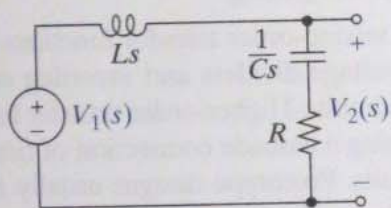


FIGURE P11-4

- 11-5 Find the driving point impedance seen by the voltage source in Figure P11-5 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

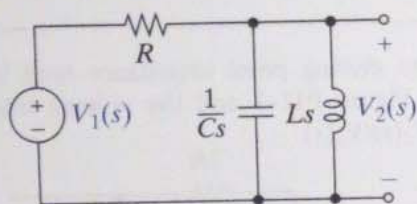


FIGURE P11-5

- 11-6 Find the driving point impedance seen by the voltage source in Figure P11-6 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

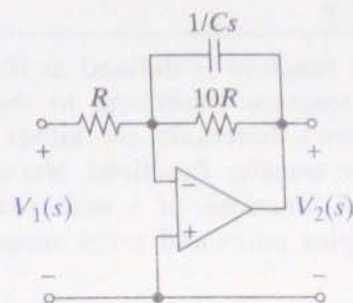


FIGURE P11-6

- 11-7 Find the driving point impedance seen by the voltage source in Figure P11-7 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

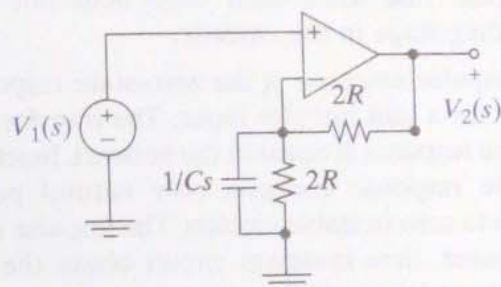


FIGURE P11-7

- 11-8 Find the driving point impedance seen by the voltage source in Figure P11-8 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

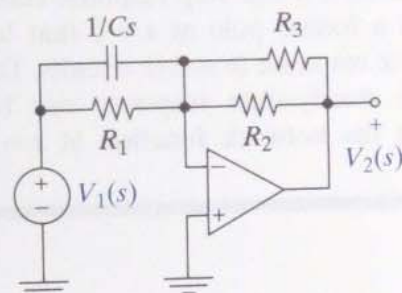


FIGURE P11-8

- 11-9 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-9.

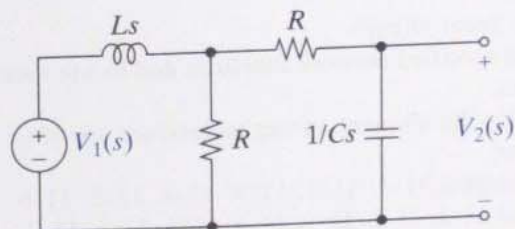


FIGURE P11-9

- 11-10 Find the driving point impedance seen by the voltage source in Figure P11-10 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$. Insert a follower at A and repeat.

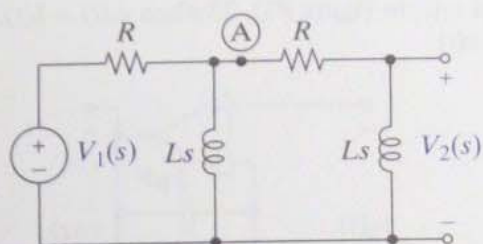


FIGURE P11-10

- 11-11 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-11. Select values of R and C so that $T_V(s)$ has a pole at $s = -100$ krad/s.

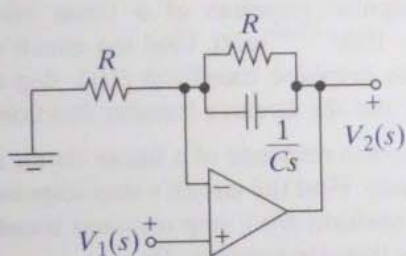


FIGURE P11-11

- 11-12 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-12. Select values of R_1 , R_2 , and C so that $T_V(s)$ has a pole at $s = -250$ krad/s and $R_2/R_1 = 100$.

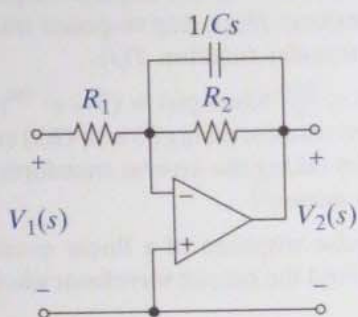


FIGURE P11-12

- 11-13 Find the current transfer function $T_I(s) = I_2(s)/I_1(s)$ in Figure P11-13. Select values of R and L so that $T_I(s)$ has a pole at $s = -377$ rad/s.

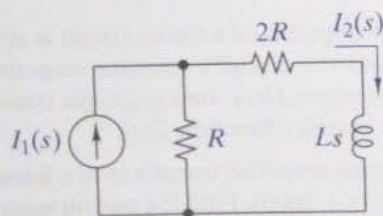


FIGURE P11-13

- 11-14 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-14. Locate the poles and zeros of the transfer function.

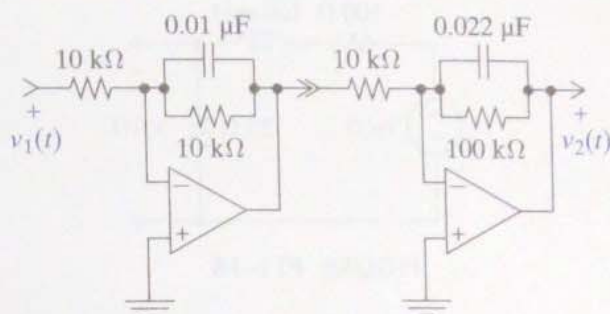


FIGURE P11-14

- 11-15 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-15. Locate the poles and zeros of the transfer function.

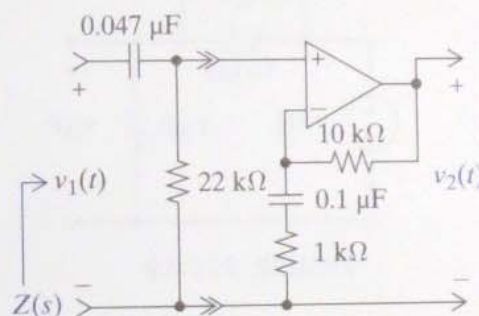


FIGURE P11-15

- 11-16 Find the input impedance $Z(s)$ in Figure P11-15.

OBJECTIVE 11-2 NETWORK FUNCTIONS, IMPULSE RESPONSE, AND STEP RESPONSES (SECTS. 11-3, 11-4)

- Given a first- or second-order linear circuit, find its impulse or step response.
- Given the impulse or step response of a linear circuit, find the network functions.
- Given the impulse or step response of a linear circuit, find the response due to other inputs.

See Examples 11-8, 11-9, 11-10, 11-11, 11-12 and Exercises 11-9, 11-10, 11-11, 11-12, 11-13, 11-14.

- 11-17 Find the impulse response at $v_2(t)$ in Figure P11-17. Find the circuit's step response.

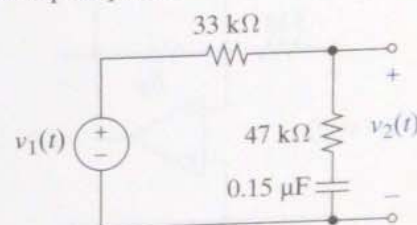


FIGURE P11-17

Laplace function to find the waveform $y(f)$ and plot the results.

- 11-33 The impulse response of a linear circuit is $h(t) = 20u(t) + \delta(t)$. Find the output waveform $y(t)$ when the input is $x(t) = 2[e^{-20t}]u(t)$.

OBJECTIVE 11-3 NETWORK FUNCTIONS AND THE SINUSOIDAL STEADY-STATE RESPONSE (SECT. 11-5)

- (a) Given a first- or second-order linear circuit with a specified input sinusoid, find the sinusoidal steady-state response.
 (b) Given the network function, impulse response, or step response, find the sinusoidal steady-state response for a specified input sinusoid.

See Examples 11-13, 11-14, 11-15 and Exercises 11-15, 11-16, 11-17, 11-18.

- 11-34 The circuit in Figure P11-34 is in the steady state with $v_1(t) = 5 \cos 1414.21t$ V. Find $v_{2SS}(t)$. Repeat for $v_1(t) = 5 \cos 1 \text{ kt}$ V. And without doing any calculations, repeat for $v_1(t) = 5$ V.

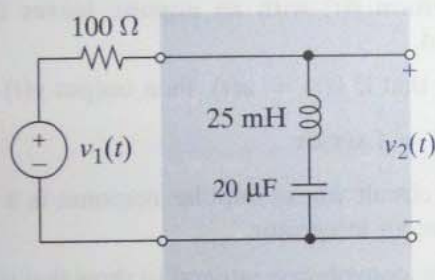


FIGURE P11-34

- 11-35 The circuit in Figure P11-35 is in the steady state with $v_1(t) = 10 \cos 500t$ V. Find $v_{2SS}(t)$. Repeat for $v_1(t) = 10 \cos 1 \text{ kt}$ V, and for $v_1(t) = 10 \cos 10 \text{ kt}$ V. Where is the pole located?

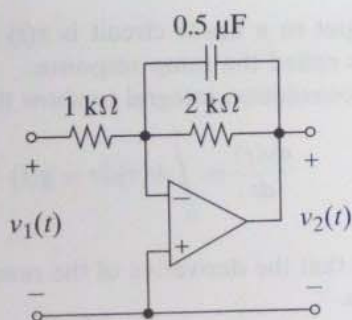


FIGURE P11-35

- 11-36 The circuit in Figure P11-36 is in the steady state with $v_1(t) = 25 \cos 2000t$ V. Find $v_{2SS}(t)$. Repeat for $v_1(t) = 25 \cos 10 \text{ kt}$ V. Where are the poles located?

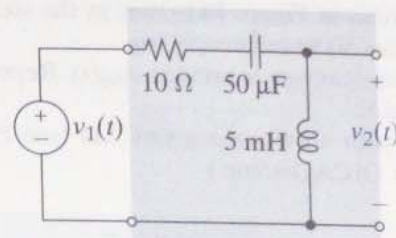


FIGURE P11-36

- 11-37 The circuit in Figure P11-37 is in the steady state with $i_1(t) = 10 \cos 50kt$ mA, $R_1 = 100 \Omega$, $R_2 = 400 \Omega$, and $L = 100 \text{ mH}$. Find $i_{2SS}(t)$. Repeat for $i_1(t) = 10 \cos 5kt$ mA. Where is the pole located?

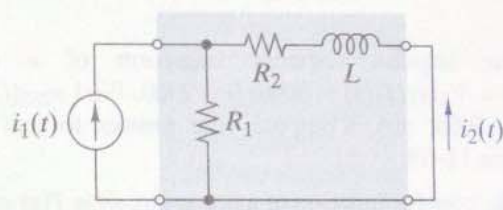


FIGURE P11-37

- 11-38 The circuit in Figure P11-38 is in the steady state with $i_1(t) = 5 \cos 1000t$ mA, $R = 1 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 0.5 \mu\text{F}$.
 (a) Find $i_{2SS}(t)$.

(b) Verify your results using OrCAD. Use IAC for a source and note that the current comes out of the negative terminal (passive sign convention.). The IPRINT element on the right side of the figure is an ammeter and it comes from the Special Library. It has a polarity as indicated by the small "-" on the element. You should ensure that it will read the correct current direction. Before you can use the IPRINT element, it needs to be set up to display the current magnitude and phase. Double click on the element and in its property editor place a "y" in AC, MAG, and PHASE boxes. Click the Apply button and close the property editor. To obtain the simulation select AC Sweep/Noise. Set the start frequency and the stop frequency at 159.15 Hz (1000 rad/s). Place a 1 in Points/Decade. Run the simulation. Under View, select Output File. Scroll down until you see the IPRINT output.

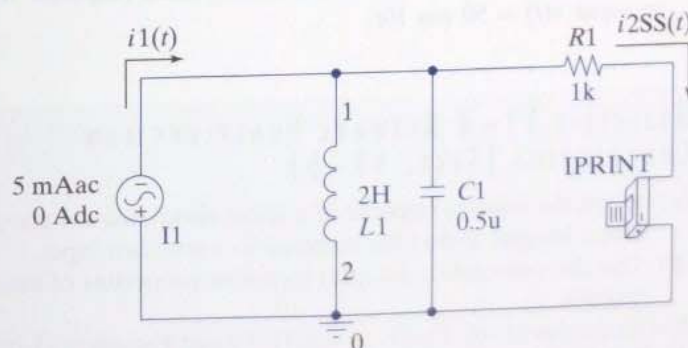


FIGURE P11-38

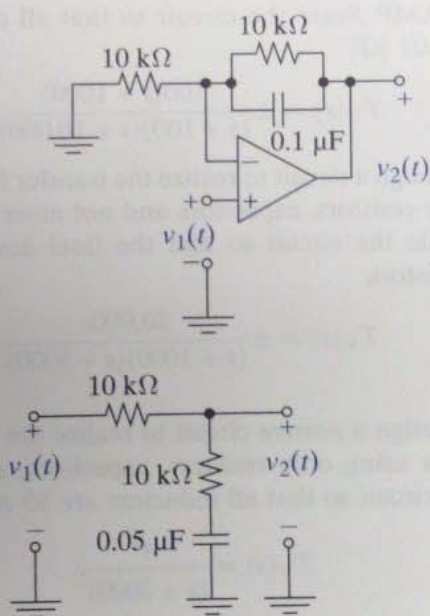


FIGURE P11-74

- 11-75 **E** It is claimed that both circuits in Figure P11-75 realize the transfer function

$$T_V(s) = \frac{\pm 1000s}{(s + 1000)(s + 4000)}$$

- (a) Verify that both circuits realize the specified $T_V(s)$.
 (b) Which circuit would you choose if the output must drive a $1 \text{ k}\Omega$ load?
 (c) Which circuit would you choose if the input comes from a 50Ω source?
 (d) It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no matter which circuit is the first stage and which is the second stage. Do you agree or disagree? Explain.

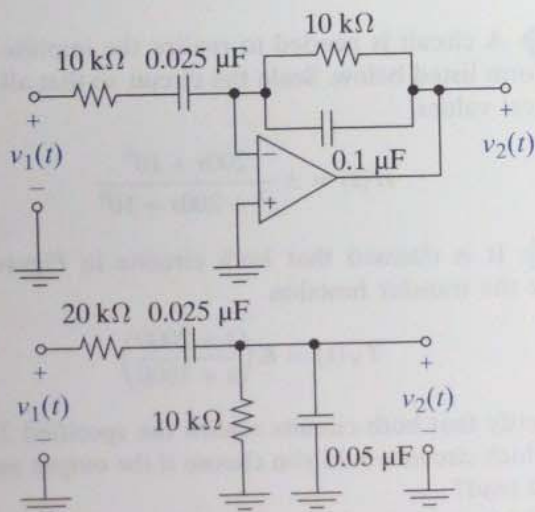


FIGURE P11-75

- 11-76 **D** Design a circuit that produces the following step response.

$$g(t) = 24[1 - e^{-50t} - 50te^{-50t}]u(t)$$

- 11-77 **D** A circuit is needed that will take an input of $v_1(t) = 25e^{-10t}u(t)$ mV and produce an output of $v_2(t) = 500e^{-200t}u(t)$ mV. Design such a circuit using practical parts values. Validate your design by using OrCAD.

- 11-78 **D** A circuit is needed that will take an input of $v_1(t) = [1 - e^{-10,000t}]u(t)$ V and produce a constant -5 V output. Design such a circuit using practical parts values. Validate your design using OrCAD.

INTEGRATING PROBLEMS

- 11-79 First-Order Circuit Impulse and Step Responses **A, D**

Each row in the table shown in Figure P11-79 refers to a first-order circuit with an impulse response $h(t)$ and a step response $g(t)$. Fill in the missing entries in the table.

Circuit	$h(t)$	$g(t)$
	$\delta(t) - [\alpha e^{-\alpha t}]u(t)$	
		$(1 + \frac{e^{-\alpha t}}{2})u(t)$

FIGURE P11-79

- 11-80 OP AMP Modules and Loading **A**

Figure P11-80 shows an interconnection of three basic OP AMP modules.

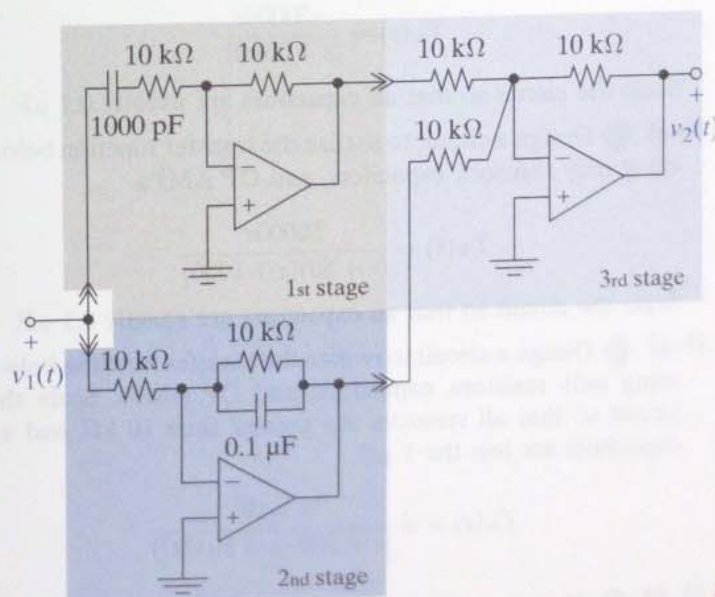


FIGURE P11-80

- (a) Does this interconnection involve loading?
- (b) Find the overall transfer function of the interconnection and locate its poles and zeros.
- (c) Find the steady-state output $v_2(t)$ when the input is $v_1(t) = \cos 500t$ V. Repeat for $v_1(t) = \cos 10kt$ V and again for $v_1(t) = \cos 200kt$ V.
- (d) Can you think of a use for this circuit?

11-81 OP AMP Modules and Stability **A**

Figure P11-81 shows an interconnection of three basic circuit modules. Does this interconnection involve loading? Find the overall transfer function of the interconnection and locate its poles and zeros. Is the circuit stable?

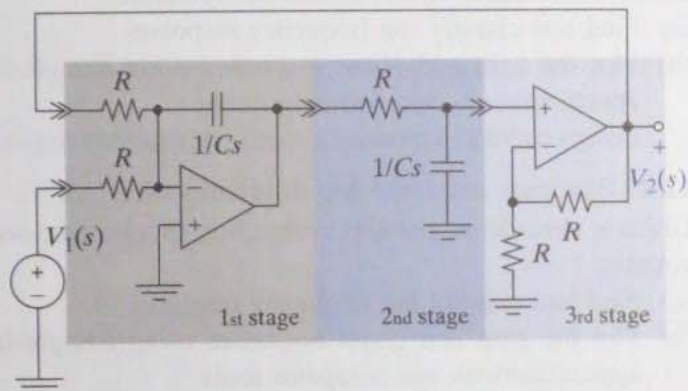


FIGURE P11-81

11-82 Step Response and Fan-Out **A**

The fan-out of a digital device is defined as the maximum number of inputs to similar devices that can be reliably driven by the device output. Figure P11-82 is a simplified

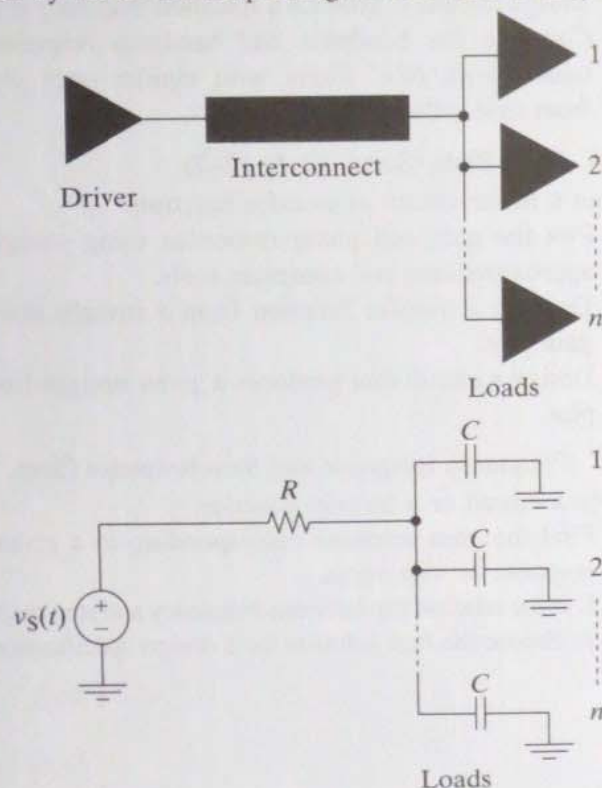


FIGURE P11-82

diagram of a device's output driving n identical capacitive inputs. To operate reliably, a 5-V step function at the device output must drive the capacitive inputs to 3.7 V in 10 ns or less. Determine the device fan-out for $R = 1\text{ k}\Omega$ and $C = 3\text{ pF}$.

11-83 Designing to Specifications **D**

A particular circuit needs to be designed that has the following transfer function requirements: Poles at $s = -100$ and $s = -10,000$; zeros at $s = 0$ and $s = -1000$; and a gain of 10 as $s \rightarrow \infty$. Find the circuit's transfer function and use MATLAB to plot its step response. Then design a circuit that will meet that requirement. Finally, use OrCAD to validate that your circuit has the same step response as found using MATLAB.

11-84 Comparison of Sinusoidal Steady-State Analysis versus Phasor Analysis **E**

A circuit designer often is faced with deciding which analysis technique to use when attempting to solve a circuit problem. In this problem we will look at the circuit in Figure P11-84 and choose which technique is the better one to use for different analysis scenarios. Explain why you selected the technique you did.

- (a) You need to calculate the circuit's transfer function $T_V(s) = V_2(s)/V_1(s)$.
- (b) The input is given as $v_1(t) = 5 \cos 1000t$ V and you need to find $v_{2SS}(t)$.
- (c) The input is given as $v_1(t) = 5 \cos 1000t$ V and you need to find $i_X(t)$.
- (d) The input is given as $v_1(t) = V_A \cos \omega t$ V and you need to find $v_{2SS}(t)$.
- (e) The input is given as $v_1(t) = 170 \cos 377t$ V and you need to find all of the voltages and currents in the circuit.
- (f) You need to find the poles and zeros of the circuit.
- (g) You need to find if the current leads or lags the voltage across the two resistors when the input is $5 \cos 1000t$ V.
- (h) You need to determine what type of filtering the circuit performs.
- (i) You need to select a load for maximum power when the input is $v_1(t) = 170 \cos 377t$ V.

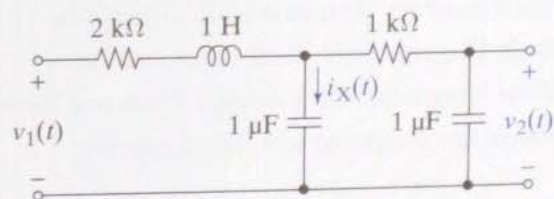


FIGURE P11-84

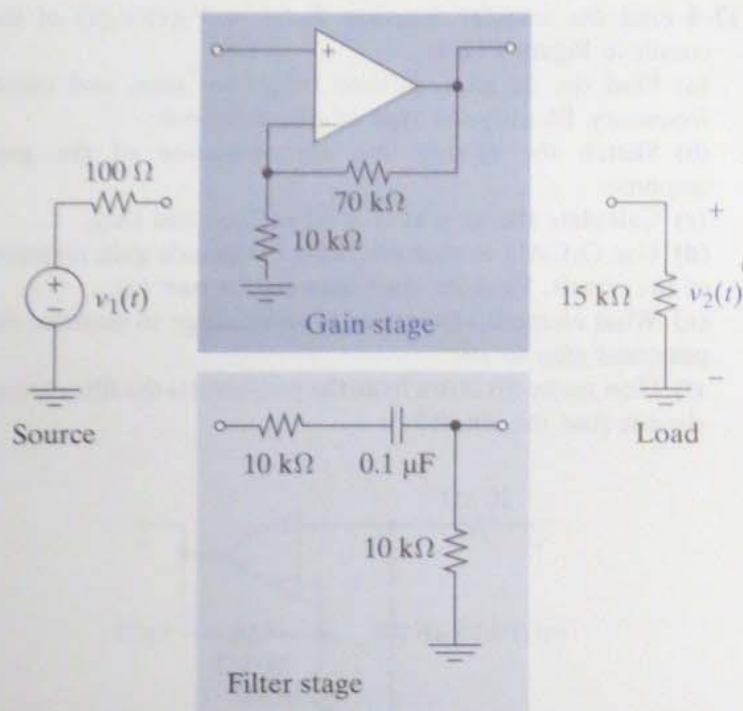


FIGURE P12-8

12-9 E A young designer needed to design a low-pass filter with a cutoff of 1 krad/s and a gain of -5 . The filter is to fit as an interface between the source and the load. The designer was perplexed when no matter how the stages shown in Figure P12-9 are connected the results are not what were expected. Explain the problem and suggest a way to achieve the desired results.

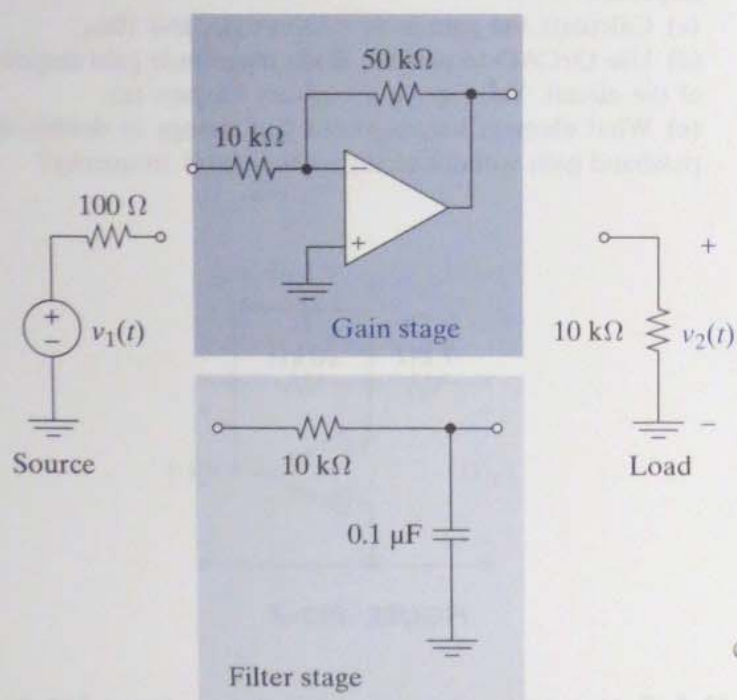


FIGURE P12-9

12-10 D Design a low-pass first-order filter with a cutoff frequency of 2000 rad/s and a passband gain of 1.

12-11 D Design an RC low-pass first-order filter with a cutoff frequency of 10,000 rad/s and a passband gain of $+10$.

12-12 D Design an RL high-pass first-order filter with a cutoff frequency of 120 Hz and a passband gain of -15 .

12-13 D Design an RC high-pass first-order filter with a cutoff frequency of 2000 rad/s and a passband gain of -100 .

12-14 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-14.

(a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.

(b) Use OrCAD to plot the Bode magnitude gain response of the circuit.

(c) What element value would you change to increase the cutoff frequency by one decade?

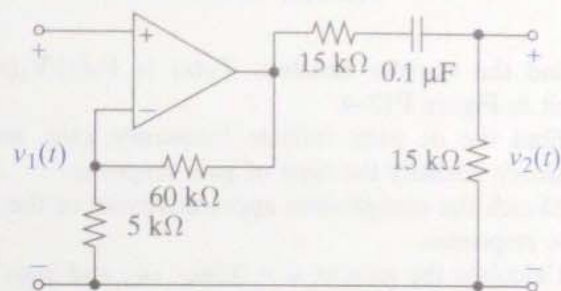


FIGURE P12-14

12-15 D Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-15. What type of gain response does the circuit have? What is the passband gain? Select practical, standard values of R and C from the inside rear cover so that the cutoff frequency is 300 kHz $\pm 5\%$.

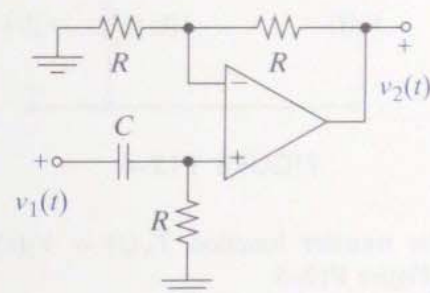


FIGURE P12-15

12-16 A first-order high-pass circuit has a passband gain of 20 dB and a cutoff frequency of 1000 rad/s. Find the gain (in dB) at $\omega = 0, 500, 1000$, and 5000 rad/s.

12-17 A first-order low-pass circuit has a passband gain of 0 dB and a cutoff frequency of 2 krad/s. Find the gain (in dB) at $\omega = 0, 200$ rad/s, 20 krad/s, and 200 krad/s.

12-18 D The transfer function of a first-order circuit is

$$T(s) = \frac{10,000}{s + 2000}$$

(a) Identify the type of gain response. Find the cutoff frequency and the passband gain.

(b) Use MATLAB to plot the magnitude of the Bode gain response.

- (c) Design a circuit to realize the transfer function.
 (d) Use OrCAD to validate your circuit design by comparing its frequency response to the MATLAB output.

12-19 D The transfer function of a first-order circuit is

$$T(s) = \frac{5s}{s + 15000}$$

- (a) Identify the type of gain response. Find the cutoff frequency and the passband gain.
 (b) Use MATLAB to plot the magnitude of the Bode gain response.
 (c) Design a circuit to realize the transfer function.
 (d) Use OrCAD to validate your circuit design by comparing its frequency response to the MATLAB output

12-20 D A student decided that she needed a low-pass filter that had a roll-off of -2 or -40dB/decade , with a cutoff frequency of 1000 rad/s . She correctly designed two identical passive RC filters, each with a cutoff of 1000 rad/s , and connected them in cascade or one after the other. She cleverly separated them with a buffer to avoid loading. When she measured the cutoff frequency she was dismayed to discover that it was not 1000 rad/s although the roll-off was correct.
 (a) Simulate the cascaded pair and determine where the actual cutoff frequency occurs.
 (b) Help the student by altering her design so that the two identical RC filters actually produce the desired outcome. Explain why the discrepancy occurred.

OBJECTIVE 12-2 BANDPASS AND BANDSTOP RESPONSES (SECT. 12-4)

Given a cascade or parallel connection of two first-order circuits:

- (a) Find and classify the frequency response.
 (b) Plot the gain and phase responses using straight-line approximations and computer tools.
 (c) Design circuits to produce a specified frequency response. See Examples 12-9, 12-10 and Exercises 12-14, 12-15, 12-16.

12-21 D The circuit in Figure P12-21 produces a bandpass response for a suitable choice of element values. Identify the elements that control the two cutoff frequencies. Select the element values so that the passband gain is 100 and the cutoff frequencies are 1000 rad/s and 40 krad/s . Use practical element values with $R \geq 10\text{ k}\Omega$ and $C \leq 1\text{ }\mu\text{F}$. Use OrCAD to validate your design.

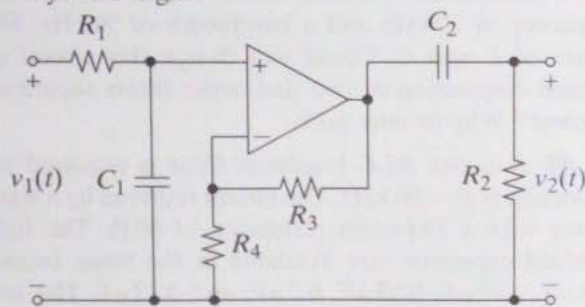


FIGURE P12-21

12-22 D Repeat Problem 12-21 for the circuit in Figure P12-22. Use OrCAD to validate your design.

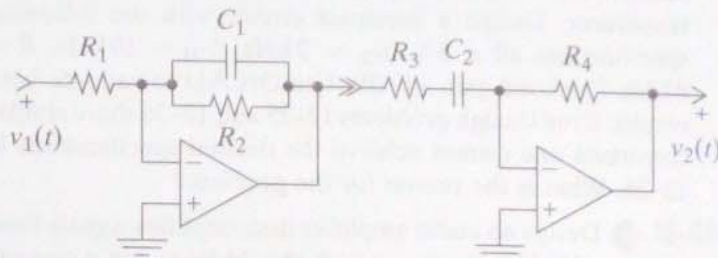


FIGURE P12-22

12-23 E Suppose the circuits you designed in problems 12-21 and 12-22 had to feed a $500\text{ }\Omega$ instrument. Which circuit would you select and why?

12-24 D The circuit in Figure P12-24 produces a bandstop response for a suitable choice of element values.

- (a) Find the circuit's transfer function.
 (b) Identify the elements that control the two cutoff frequencies. Select the element values so that the cutoff frequencies are 200 krad/s and 4000 krad/s . Use practical element values with $R_s \geq 10\text{ k}\Omega$ and $C_s \leq 1\text{ }\mu\text{F}$. Design your passband gain to be $+20\text{ dB}$.
 (c) Use MATLAB to plot the Bode magnitude plot for the values you selected.
 (d) Simulate your circuit using OrCAD and compare the results to the MATLAB output.

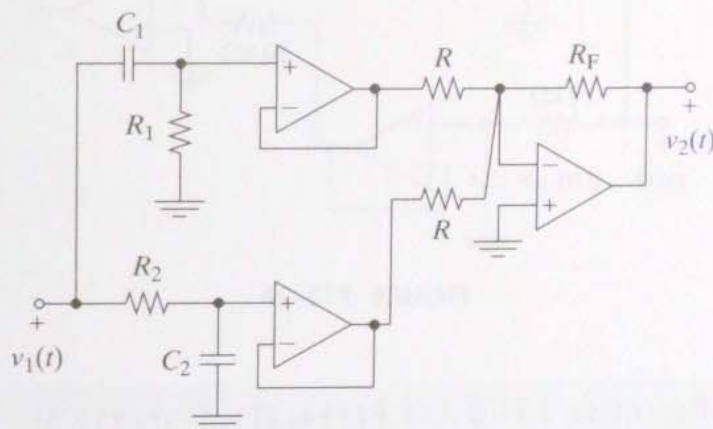


FIGURE P12-24

12-25 D Design an audio amplifier that amplifies signals from 20 Hz to 20 kHz . Your approach should be to use a cascade connection of two first-order passive circuits separated by a non-inverting OP AMP. The source has a $50\text{-}\Omega$ series resistor and the output of the filter feeds a $10\text{-k}\Omega$ audio transducer. Design a bandpass circuit with the following specifications all $\pm 5\%$: $f_{CL} = 20\text{ Hz}$, $f_{CH} = 20\text{ kHz}$, $B = 19980\text{ Hz}$. Passband gain $+40\text{ dB}$. Use OrCAD to validate your results.

12-26 D E Design an audio amplifier that amplifies signals from 2 kHz to 10 kHz . Your approach should be to use a